

Contoured Rocket Nozzles

By

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(With 6 Figures)

Abstract — Zusammenfassung — Résumé

Contoured Rocket Nozzles. A method of designing rocket nozzle wall contours to yield optimum thrust is presented. The rocket exhaust gases are treated under the assumption of isentropic, adiabatic and frictionless flow. Nozzle length and ambient pressure appear as governing restrictions under which the thrust is maximized. The solution of this extremal problem yields flow properties required along a suitably chosen control surface, and the nozzle contour is constructed by the method of characteristics to yield this required flow field. A typical numerical example is given. For the propellant combination used in the example, it is inferred from available experimental data that the expanding exhaust gases are in chemical equilibrium. The varying gas properties associated with chemical equilibrium are accounted for in the thrust optimization method. The low altitude performance of such a contoured nozzle is discussed.

Profilerte Raketendüsen. Es wird eine Methode zum Entwurf der Profile von Raketendüsenwänden für optimalen Schub beschrieben. Dabei wird angenommen, daß die aus der Rakete ausströmenden Gase einen isentropischen, adiabatischen und reibungsfreien Fluß aufweisen. Die Länge der Düse und der in der Umgebung herrschende Druck treten als beherrschende Einschränkungsfaktoren auf, unter deren Wirkung der Schub den größten Wert erreichen kann. Die Lösung dieses extremen Problems erfordert Eigenschaften des notwendigen Flusses entlang einer geeignet gewählten Kontrolloberfläche; das Düsenprofil wird nach der Charakteristikenmethode konstruiert, um diesen erforderlichen Fluß zu liefern. Dafür wird ein typisches numerisches Beispiel angegeben. Für die dabei benutzte Treibstoffkombination kann man aus verfügbaren experimentellen Daten folgern, daß die expandierenden ausströmenden Gase sich im chemischen Gleichgewicht befinden. Die mit dem chemischen Gleichgewicht verbundenen variierenden Gaseigenschaften werden bei der Methode der „Schuboptimalisierung“ berücksichtigt. Es wird die Leistung einer solchen profilierten Düse bei niedriger Höhe erörtert.

Tuyères profilées pour fusées. Présentation d'une méthode pour le dessin du profil d'une tuyère donnant une poussée optimale. L'écoulement des produits de combustion est supposé isentropique et adiabatique. La longueur de la tuyère et la pression ambiante apparaissent comme des limitations dans le problème de l'optimisation. Sa solution donne les propriétés de l'écoulement le long d'une surface de contrôle convenablement choisie et le profil s'en déduit par la méthode des caractéristiques. Un exemple numérique typique est donné. Dès données expérimentales existantes on peut inférer que les produits de combustion, correspondant au mélange d'ergols choisi,

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sont en équilibre thermodynamique. Les variations de propriété du gaz associées à l'équilibre chimique sont prises en considération dans la méthode d'optimisation. Les performances de basse altitude de telles tuyères sont mises en discussion.

Nomenclature

A = Cross section area of nozzle	x = Coordinate measured along nozzle axis
a = Local speed of sound	y = Radial coordinate
F = Thrust of nozzle	α = MACH angle
F_i = Thrust of ideal nozzle, i.e., full expansion to ambient pressure	γ = Isentropic exponent
g = Gravitational acceleration 32.17 ft/sec ²	ρ = Density
L = Nozzle length	Φ = Inclination of control surface to nozzle axis
M = MACH number	θ = Flow direction measured from nozzle axis
$M^* = W/a_i$	
P_c = Chamber pressure	
p = Static pressure	
R_t = Radial coordinate at throat	
T = Temperature	
W = Velocity	

Subscripts

a	= ambient
E	= at nozzle exit

Summary

Rocket exhaust nozzles suffer thrust loss due to flow divergence from the axial direction. A method for designing the wall contour of an exhaust nozzle to yield optimum thrust is formulated using the nozzle length, ambient pressure, and flow conditions in the throat region, as governing conditions under which the thrust is maximized. The optimum thrust nozzle contours thus obtained depend greatly upon the gas properties used for the expansion process.

At the high temperatures occurring in the rocket nozzles, chemical reactions take place during the expansion process, resulting in varying gas properties along the length of the nozzle. Experimental temperature measurements taken at the exit of a rocket nozzle indicate the possibility of achieving chemical equilibrium during the expansion process in the nozzle. If one assumes that the exhaust gases are always in chemical equilibrium, the method of characteristics for the analysis of nozzle flows can still be easily applied. A typical nozzle contour designed for optimum thrust assuming chemical equilibrium is presented and the thrust advantage is discussed.

The nozzle wall pressure at the exit of an optimum thrust contour is considerably higher than that for a conical nozzle. Accordingly, when flow separation in the nozzle is taken into account, the low altitude performance of an optimum thrust contour differs considerably from that of a conical nozzle. Typical thrust performance curves are included showing these differences over a range of altitudes.

Introduction

The thrust produced by the diverging portion of an exhaust nozzle is of considerable importance for engines operating at high altitudes. Maximum possible thrust can be obtained by expanding the exhaust gases to ambient pressure through a nozzle designed to give a parallel uniform exit flow. FOELSCH's Method [1] is suitable to compute such contours. At very low ambient pressures, even the shortest nozzle designed by this method would be excessively long and heavy.